Per-site ΔRMSE from adding mortality — C+T vs C+T+M

Below 0 = mortality reduces the site's RMSE | bars coloured by total mortality class (site maximum) | repeated CV (3×80 %) | tree cover ≥30 % (101 sites, 361 site-years) · deadtrees v5

XGBoost (Optuna) · 12m | **XGBoost (Optuna) · 24m** | **LightGBM (Optuna) · 12m** | **LightGBM (Optuna) · 24m**

ΔRMSE = RMSE(with M) − RMSE(without M)

GPPsat (µmol m⁻² s⁻¹)

53% improved | 59% improved | 56% improved | 56% improved

2 | 0 | -2

NEPmax (µmol m⁻² s⁻¹)

72% improved | 69% improved | 68% improved | 69% improved

2 | 0 | -2

ETmax (mm d⁻¹)

62% improved | 54% improved | 57% improved | 57% improved

0.015 | 0.010 | 0.005 | 0.000 | -0.005 | -0.010 | -0.015

WUE (g C mm⁻¹)

50% improved | 54% improved | 52% improved | 50% improved

0.4 | 0.0 | -0.4

Sites (sorted by ΔRMSE within each panel)

Total mortality
site maximum, mean of 100-500 m buffers
(breaks: mean ± 0.5 SD)

- Low (< 10.6 %, 41 sites)
- Mid (10.6–24.0 %, 38 sites)
- High (> 24.0 %, 22 sites)